MODULE-WISE QUESTION BANK

M1: Introduction to Networking

1. Define networking. What is the importance of networking?
2. Explain the major components of a computer network.
3. What are LAN, MAN, and WAN? Compare them.
4. Describe any two network topologies.
5. Explain Star, Ring, Bus, and Mesh topologies with diagrams.
6. Explain the OSI reference model and its 7 layers.
7. Explain the TCP/IP reference model.
8. Compare the OSI and TCP/IP reference models.

M2: Physical Layer

1. Explain various data transmission concepts.
2. What is guided media? Explain twisted pair, coaxial cable, and optical fiber.
3. What is unguided media? Explain radio, VHF, microwave, satellite, and infrared.
4. Explain NRZ, Manchester, and Differential Manchester encoding.
5. Explain FDM, TDM, and WDM multiplexing techniques.
6. Differentiate between circuit switching, packet switching, and message switching.
7. Explain with diagrams the different encoding schemes.

M3: Data Link Layer

1. What are the issues handled by the Data Link Layer?
2. Explain Stop-and-Wait flow control.
3. Explain Sliding Window flow control.
4. Explain parity check, CRC, and checksum.
5. Explain Hamming code with an example.
6. Explain Stop-and-Wait ARQ, Go-Back-N ARQ, and Selective Reject ARQ.
7. What is HDLC? Explain its frame format.
8. Explain ALOHA and Slotted ALOHA.
9. Explain collision-free protocols.
10. Write short notes on IEEE 802.3, IEEE 802.4, and IEEE 802.5.

M4: Network Layer

1. Explain the shortest path routing algorithm.
2. Explain flow-based routing.
3. Explain distance vector routing.
4. Explain broadcast and multicast routing.
5. Explain IPv4 addressing with classes.
6. Describe IPv6 features and structure.
7. Explain subnetting with an example.
8. What is CIDR? Explain with example.
9. Explain RIP, OSPF, and BGP routing protocols.
10. What is NAT? Explain its types.
11. What is ICMP? Explain its functions.

M5: Transport Layer

1. Explain the functions of the transport layer.
2. Explain TCP's 3-way handshake.
3. Explain TCP's 4-way connection termination.
4. Compare TCP and UDP.
5. What is port addressing? Explain with examples.
6. Explain leaky bucket congestion control.
7. Explain token bucket congestion control.

M6: Application Layer

1. Explain DNS and its working.
2. Explain HTTP and HTTPS.
3. Explain FTP architecture and working.
4. Explain SMTP, POP3, and IMAP.
5. What is Telnet? Explain remote login.
6. What is SSH? How is it different from Telnet?
7. Explain NFS and SMB file-sharing protocols.

8. Explain the role of a domain name server.
9. Explain the Simple Network Management Protocol (SNMP).

M7: Emerging Networking Concepts

1. What is cloud networking? Explain its characteristics.
2. Explain IoT networking architecture.
3. Explain MQTT protocol and its working.
4. Explain CoAP protocol features.
5. What is Software-Defined Networking (SDN)?
6. Explain the SDN architecture (Control plane + Data plane).
7. Compare traditional networking and SDN.